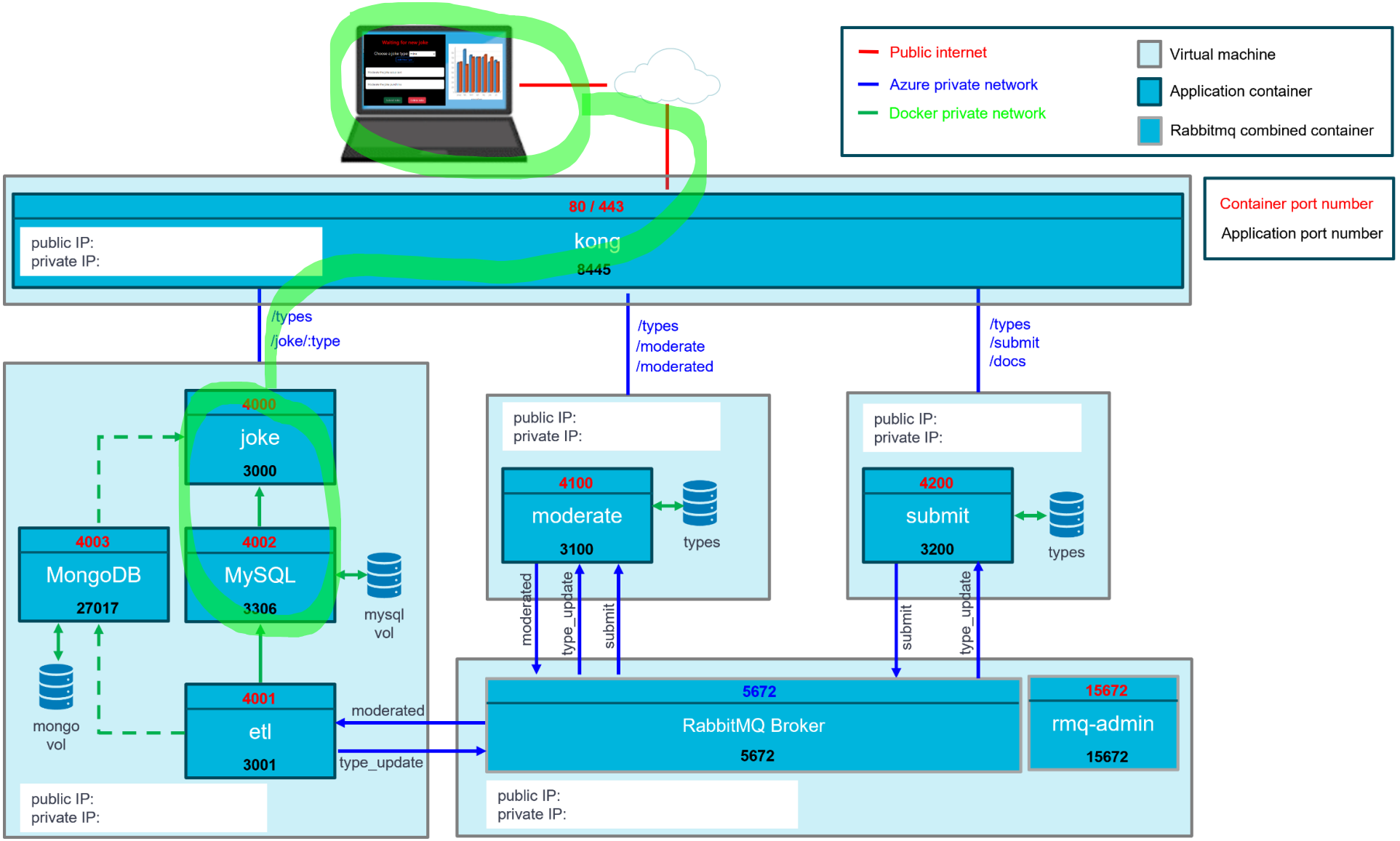
Lab 4: Add middleware to the app server and add a database



Lab 4: Implement promises, middleware and a database

Exercise 1: promises and middleware

Overview

When your joke API receives a request, you want to collect analytics information such as:

- Which joke types are most popular
- What time of day traffic is highest
- How many jokes users request per call
- Which IP addresses are making requests

You will build middleware that logs each request to a JSON file. That file could then be ingested by business analytics software to monitor the effectiveness of the service (you don't need to do this).

Your log entries should each look like the following:

```
{
  "ip": "127.0.0.1",
  "time": "2026-10-10T20:43:00.123Z",
  "type": "dad",
  "limit": 5
}
```

Requirements

1. Middleware should run only for the **GET /jokes** endpoint
   To get the timestamp you can use new Date().toISOString()
   Other information is in the req object for you to extract
2. Read the existing JSON log file into an array
   Append data to the array
   Save the updated array back to the file
3. Although the fs library has introduced promise-based versions, you should use the original callback version, wrap it in a promise then call functions of your own called **readLogFile** and **writeLogFile**
   If the read and write are successful, resolve the promise, if there is an error, reject the promise
   Test for promise success and failure
4. Test with variations of call - i.e. different joke types, different count etc.

Your log file should look something like:



Exercise 2: add a database to your jokes app

Based on the demo of basic_db, improve your jokes solution by getting your data from a database instead of a JSON file.

Download **mySQL server 8.4** from and workbench from **https://dev.mysql.com/downloads/mysql/**

When installing, select the typical option

Leave the mysql configurator selected then next and follow the instructions

For windows users the .msi is the easiest installation approach

If it asks you to create an account, ignore it and find the smaller writing saying "No thanks, just start download"

and workbench **8.0** from **https://dev.mysql.com/downloads/mysql**

I've created a video **download mysql server workbench and create a simple database.mp4** to help you with this and your lab exercise

Follow the tutorial and demo to create a database and tables using workbench

Create a database with two tables, one for the jokes and one for types. For basic operation and testing, manually add a about six jokes and a couple of types.

Update your app server to connect to your database and return data using your existing endpoints. You can write the SQL directly into the routes for now just to get started as shown in the basic_db demo and video.

You should see the data from the types table populate the types dropdown in the front end and the jokes setup and punchline returned from the jokes table and displayed in the same way it was previously.

You don’t need the tables to be related at this point - i.e. you can duplicate the types in the jokes table to search on as this is just an integration test, we will look at databases in a bit more detail later including forming relationships and referential integrity.

If however, you have already studied databases, feel free to implement the relationships, joins and referential integrity but I'll be going through that in the next session which is on databases.